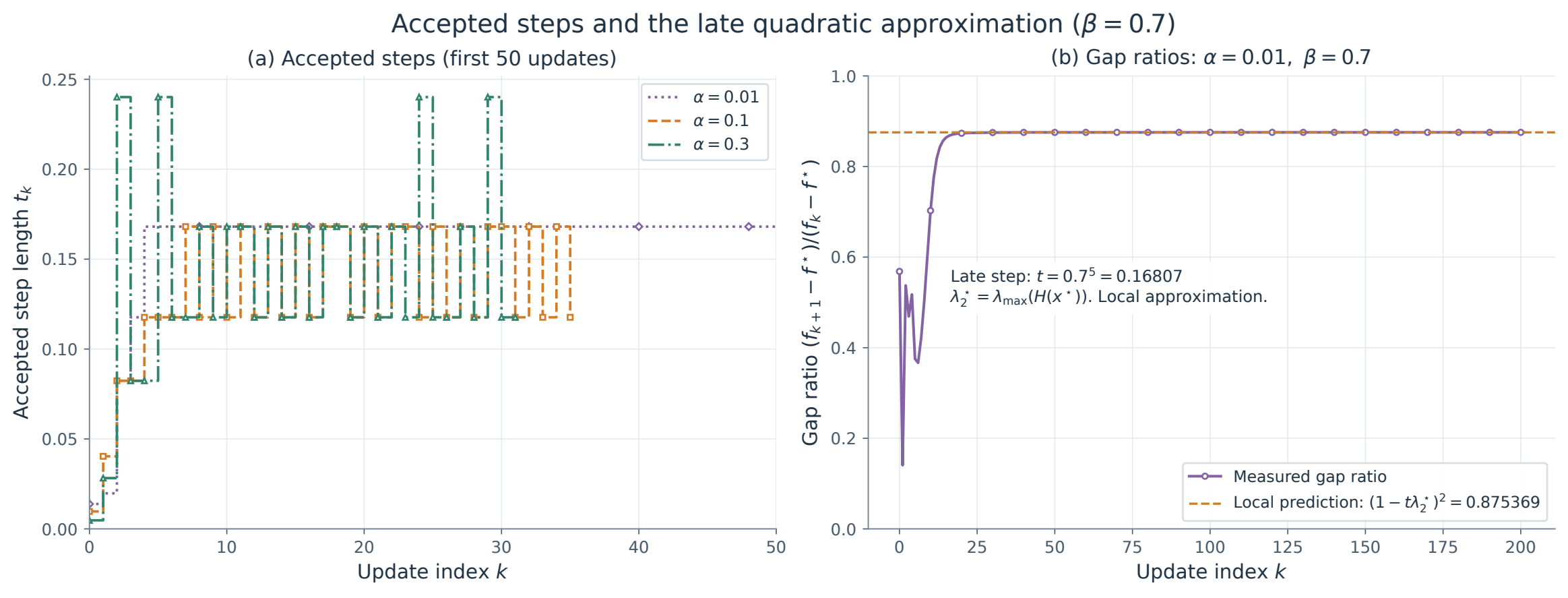
Accepted steps and the late quadratic approximation ($\beta = 0.7$)
(a) Accepted steps (first 50 updates)
Accepted step length $t_k$
Update index $k$
$\alpha = 0.01$
$\alpha = 0.1$
$\alpha = 0.3$
(b) Gap ratios: $\alpha = 0.01,\ \beta = 0.7$
Gap ratio $(f_{k+1} - f^\star)/(f_k - f^\star)$
Late step: $t = 0.7^5 = 0.16807$
$\lambda_2^\star = \lambda_{\max}(H(x^\star))$. Local approximation.
Measured gap ratio
Local prediction: $(1 - t\lambda_2^\star)^2 = 0.875369$
Update index $k$